

Analysis

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Key Takeaways

- The dataset includes **215 cargo movements** from **17 port authorities**, **69 facilities** and **180 different companies** with approximately 109 million metric tons of hazardous cargo reported in 2022. Geographic concentration is striking **İzmir (Aliğa), Adana (Ceyhan)** and **Hatay (İskenderun)** dominate the volume while most other provinces handle only a sliver of the total.
- **Company-level tonnage is deeply uneven.** The bulk of receivers sit in the 10³–10⁴ ton range yet a handful of operators push well past 10⁵ and that gap matters enormously for policy. Just **107 of the 180 companies** clear the 20,000 ton threshold but together they move roughly **99.5% of all hazardous cargo**. Wide by headcount, narrow by liability.
- A hypothetical **50/ton levy** on those threshold crossing companies would bring in around **5.4 billion** most of it coming from the top ten handlers alone.

1. Overview of the dataset

```
overview <- hns2022 |>
  summarise(
    n_records      = n(),
    n_ports        = n_distinct(port_authority_short),
    n_facilities   = n_distinct(facility),
    n_companies    = n_distinct(company_clean),
    total_cargo_mt = sum(cargo_mt)
  )
overview
```

```
# A tibble: 1 x 5
  n_records n_ports n_facilities n_companies total_cargo_mt
  <int>    <int>    <int>    <int>    <dbl>
1      215      17         69       180    108894605.

215 Records
17 Port authorities
69 Facilities
```

180 Companies (cleaned)

108.9 M ton Total 2022 cargo

The dataset is a long table where each row pairs a **receiver company** with the **facility** that handled their cargo and the **regional port authority** that oversees that facility. The numeric column `cargo_mt` is what we'll be slicing in every direction below.

2. Where is hazardous cargo concentrated?

Each port authority is mapped to its host province, and 2022 tonnage is summed accordingly. The darker the province, the heavier the load.

```
prov_totals <- hns2022 |>
  group_by(iso2, province) |>
  summarise(total_mt = sum(cargo_mt), .groups = "drop")

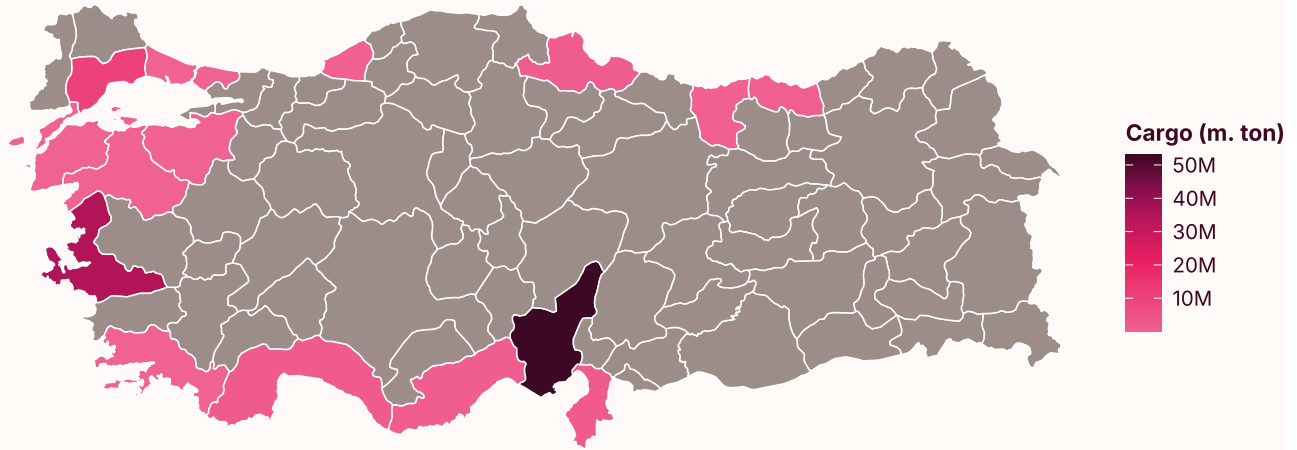
tr_states <- ne_states(country = "Turkey", returnclass = "sf")

if ("iso_3166_2" %in% names(tr_states)) {
  tr_map <- tr_states |>
  left_join(prov_totals, by = c("iso_3166_2" = "iso2"))
} else {
  norm <- function(x) tolower(stringi::stri_trans_general(x, "Any-Latin; Latin-ASCII"))
  tr_map <- tr_states |>
  mutate(name_n = norm(name)) |>
  left_join(prov_totals |> mutate(name_n = norm(province)), by = "name_n")
}

ggplot(tr_map) +
  geom_sf(aes(fill = total_mt), color = "white", linewidth = 0.35) +
  scale_fill_gradientn(
    colours = c("#F06292", "#EC407A", "#E91E63", "#C2185B", "#880E4F", "#3D0823"),
    na.value = "#9C8D88",
    labels = scales::label_number(scale_cut = scales::cut_short_scale()),
    name = "Cargo (m. ton)"
  ) +
  coord_sf(crs = 4326) +
  labs(
    title = "Türkiye's hazardous cargo, 2022",
    subtitle = "Provinces are filled by the total tonnage handled at port authorities they host. Grey = no",
    caption = "Source: Directorate General for Maritime Affairs, 2022. Author calculations."
  ) +
  theme_void(base_family = "Inter") +
  theme(
    plot.title = element_text(family = "Playfair Display", face = "italic", size = 22,
                              color = "#3D0823", margin = margin(b = 6)),
    plot.subtitle = element_text(color = "#3D0823", size = 12, margin = margin(b = 10)),
    plot.caption = element_text(color = "#3D0823", size = 10, hjust = 0,
                                margin = margin(t = 8)),
    legend.title = element_text(color = "#3D0823", face = "bold", size = 11),
    legend.text = element_text(color = "#3D0823", size = 10),
    legend.position = "right",
    plot.background = element_rect(fill = "#FFFAF7", color = NA),
    panel.background = element_rect(fill = "#FFFAF7", color = NA)
  )
```

Türkiye's hazardous cargo, 2022

Provinces are filled by the total tonnage handled at port authorities they host. Grey = no port authority in that provi



Source: Directorate General for Maritime Affairs, 2022. Author calculations.

Figure 1: Total hazardous cargo handled in 2022, by host province of the port authority. The darker the province, the heavier the load.

The Aegean coast dominates **İzmir** alone hosts both the **Aliğa** and **İzmir** port authorities, making it the single heaviest concentration on the map. The Eastern Mediterranean follows close behind with **Adana's Ceyhan** terminals and **Hatay's İskenderun** accounting for much of the remainder. The Black Sea, Marmara and Antalya regions are present but noticeably lighter.

3. How is cargo distributed across companies?

Each company's 2022 tonnage is summed and plotted on a log scale to capture the full spread from the smallest handlers to the largest:

```
company_totals <- hns2022 |>
  group_by(company_clean) |>
  summarise(total_mt = sum(cargo_mt), .groups = "drop") |>
  filter(total_mt > 0)

# Get the density curve's y-max so the threshold label can sit at a fixed
# fraction of the curve's height. This is more reliable than y = Inf with
# vjust on a log-x axis.
density_y_max <- max(
  ggplot_build(
    ggplot(company_totals, aes(x = total_mt)) +
      geom_density() +
      scale_x_log10()
  )$data[[1]]$y
)

ggplot(company_totals, aes(x = total_mt)) +
```

```

geom_density(fill = "#FF80B0", alpha = 0.55, color = "#C2185B", linewidth = 1) +
geom_vline(xintercept = 20000, linetype = "dashed", color = "#00C9D8", linewidth = 1) +
annotate(
  "text",
  x = 20000, y = density_y_max * 0.92,
  label = "20,000-ton\nthreshold",
  hjust = -0.08, vjust = 1,
  color = "#00838F", fontface = "bold",
  size = 3.8, family = "Inter"
) +
scale_x_log10(labels = scales::label_number(scale_cut = scales::cut_short_scale())) +
labs(
  title = "Most companies are small. A handful are enormous.",
  subtitle = "Density of company-level total tonnage in 2022 (log-10 axis).",
  x = "Total cargo per company (metric tons, log scale)",
  y = "Density",
  caption = "Each company aggregated by its cleaned, deduplicated name."
) +
theme_riyo()

```

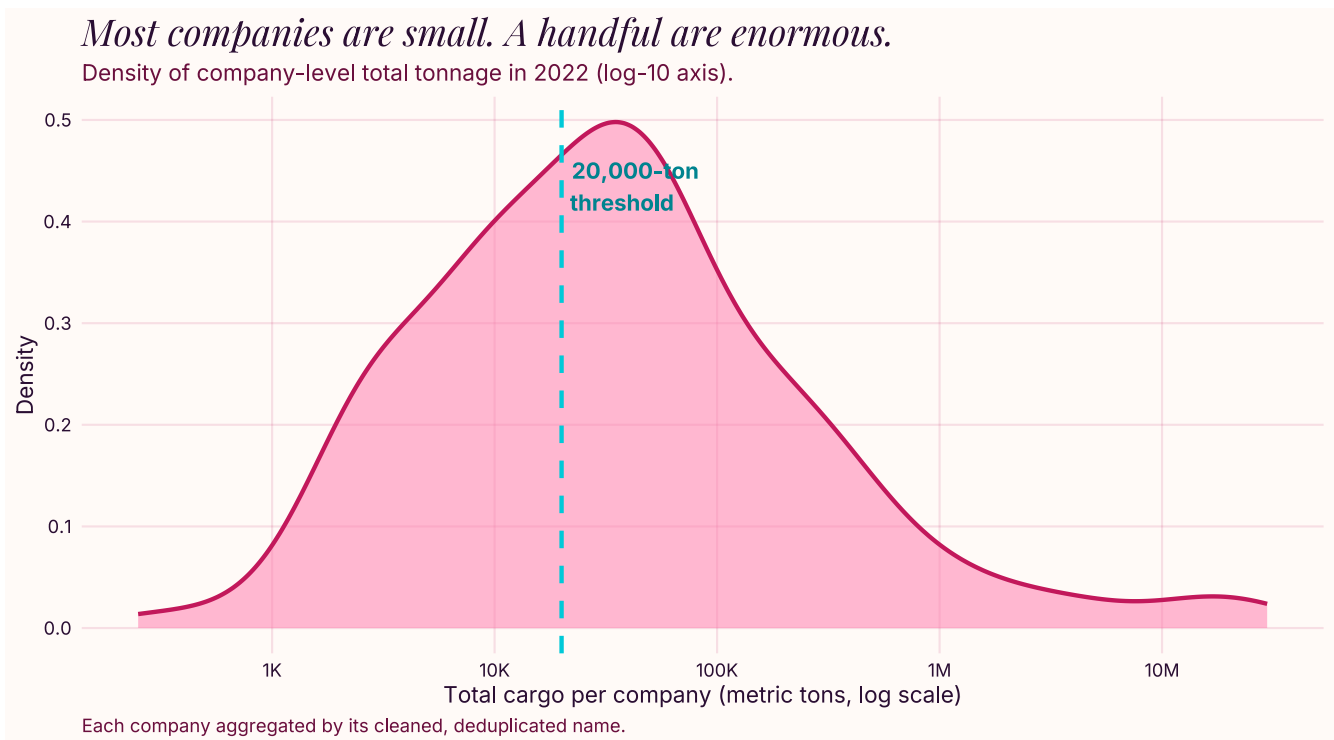


Figure 2: Density of total 2022 hazardous cargo tonnage per company (log scale). The cyan dashed line is the 20,000 ton policy threshold.

The pattern is hard to miss. Most companies cluster in the **10³–10⁴ ton** range, but a thin upper tail stretches well past **10⁵ tons** and that tail is where the real volume lives. On a linear axis, the smaller handlers would barely register; they outnumber the giants but are dwarfed by them in every other sense. ## 4. What would a tonnage levy actually cost? (Hypothetical simulation)

The Directorate is considering tax/regulatory measures targeting companies that exceed **20,000 tons** of hazardous cargo per year. To put a number on what that would mean, we apply a hypothetical **50 per ton** levy on the *full* tonnage of any company that crosses the threshold.

Disclaimer. 50/ton is illustrative — chosen for arithmetic clarity, not based on any draft policy. Treat absolute figures as proportions, not forecasts.

```
TAX_RATE_TL_PER_TON <- 50

co_tax <- company_totals |>
  mutate(
    above_threshold = total_mt > 20000,
    tax_TL = if_else(above_threshold, total_mt * TAX_RATE_TL_PER_TON, 0)
  )

threshold_summary <- co_tax |>
  summarise(
    n_total      = n(),
    n_taxed      = sum(above_threshold),
    pct_companies = mean(above_threshold) * 100,
    total_mt     = sum(total_mt),
    taxed_mt     = sum(total_mt[above_threshold]),
    pct_taxed_mt = sum(total_mt[above_threshold]) / sum(total_mt) * 100,
    total_tax_TL = sum(tax_TL),
    total_tax_billion = sum(tax_TL) / 1e9
  )

threshold_summary

# A tibble: 1 x 8
  n_total n_taxed pct_companies total_mt taxed_mt pct_taxed_mt total_tax_TL
  <int>   <int>   <dbl>       <dbl>   <dbl>       <dbl>       <dbl>
1    180    107     59.4 108894605.    NA         NA    5416538302.
# i 1 more variable: total_tax_billion <dbl>

share_df <- tibble(
  metric = factor(
    c("Companies", "Tonnage moved"),
    levels = c("Companies", "Tonnage moved")
  ),
  above = c(threshold_summary$pct_companies, threshold_summary$pct_taxed_mt),
  below = c(100 - threshold_summary$pct_companies, 100 - threshold_summary$pct_taxed_mt)
) |>
  pivot_longer(c(above, below), names_to = "side", values_to = "pct") |>
  mutate(side = factor(side, levels = c("below", "above"),
    labels = c("Below 20k tons", "Above 20k tons")))

ggplot(share_df, aes(x = metric, y = pct, fill = side)) +
  geom_col(width = 0.55) +
  geom_text(
    aes(label = sprintf("%.0f%%", pct)),
    position = position_stack(vjust = 0.5),
    color = "white", fontface = "bold", size = 5,
    family = "Inter"
  ) +
  scale_fill_manual(values = c("Below 20k tons" = "#FFC1D8", "Above 20k tons" = "#C2185B")) +
  coord_flip() +
  labs(
    title = "Wide net, top-heavy bill.",
    subtitle = sprintf(
      "%.0f%% of companies cross the 20,000-ton threshold - and together they account for ~%.1f%% of all h
      threshold_summary$pct_companies, threshold_summary$pct_taxed_mt
    )
  )
```

```

),
x = NULL, y = "Share (%)", fill = NULL,
caption = "50/ton hypothetical levy applied to crossing companies only."
) +
theme_riyo() +
theme(legend.position = "top")

```

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_col()`).

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_text()`).

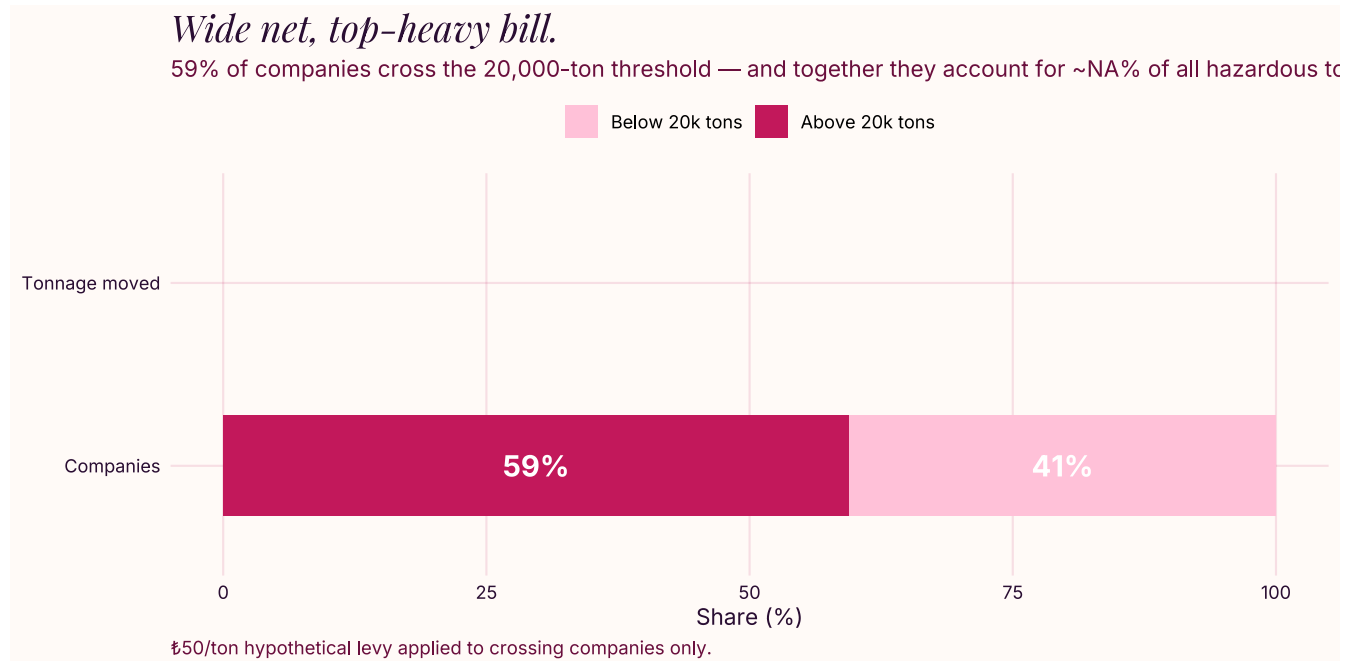


Figure 3: How many companies exceed the threshold — and what share of total tonnage they account for.

```

top10_tax <- co_tax |>
  filter(above_threshold) |>
  slice_max(tax_TL, n = 10) |>
  mutate(
    label_short = str_trunc(company_clean, 38),
    label_short = fct_reorder(label_short, tax_TL)
  )

ggplot(top10_tax, aes(x = label_short, y = tax_TL / 1e6)) +
  geom_col(fill = "#C2185B", width = 0.7) +
  geom_text(
    aes(label = scales::label_number(accuracy = 1, big.mark = ",")(tax_TL / 1e6)),
    hjust = -0.15, color = "#2A0E33", family = "Inter", fontface = "bold"
  ) +
  coord_flip() +
  scale_y_continuous(
    labels = scales::label_number(suffix = "M "),
    expand = expansion(mult = c(0, 0.18))
  ) +

```

```
labs(
  title = "Where the bill would land - top 10 hypothetical taxpayers",
  subtitle = sprintf(
    "Total industry liability %.1f billion · top 10 share %.0f%%",
    threshold_summary$total_tax_billion,
    sum(top10_tax$tax_TL) / threshold_summary$total_tax_TL * 100
  ),
  x = NULL, y = "Hypothetical tax (million )",
  caption = "Hypothetical levy: 50 per ton on companies above 20,000 tons of 2022 hazardous cargo."
) +
theme_riyo()
```

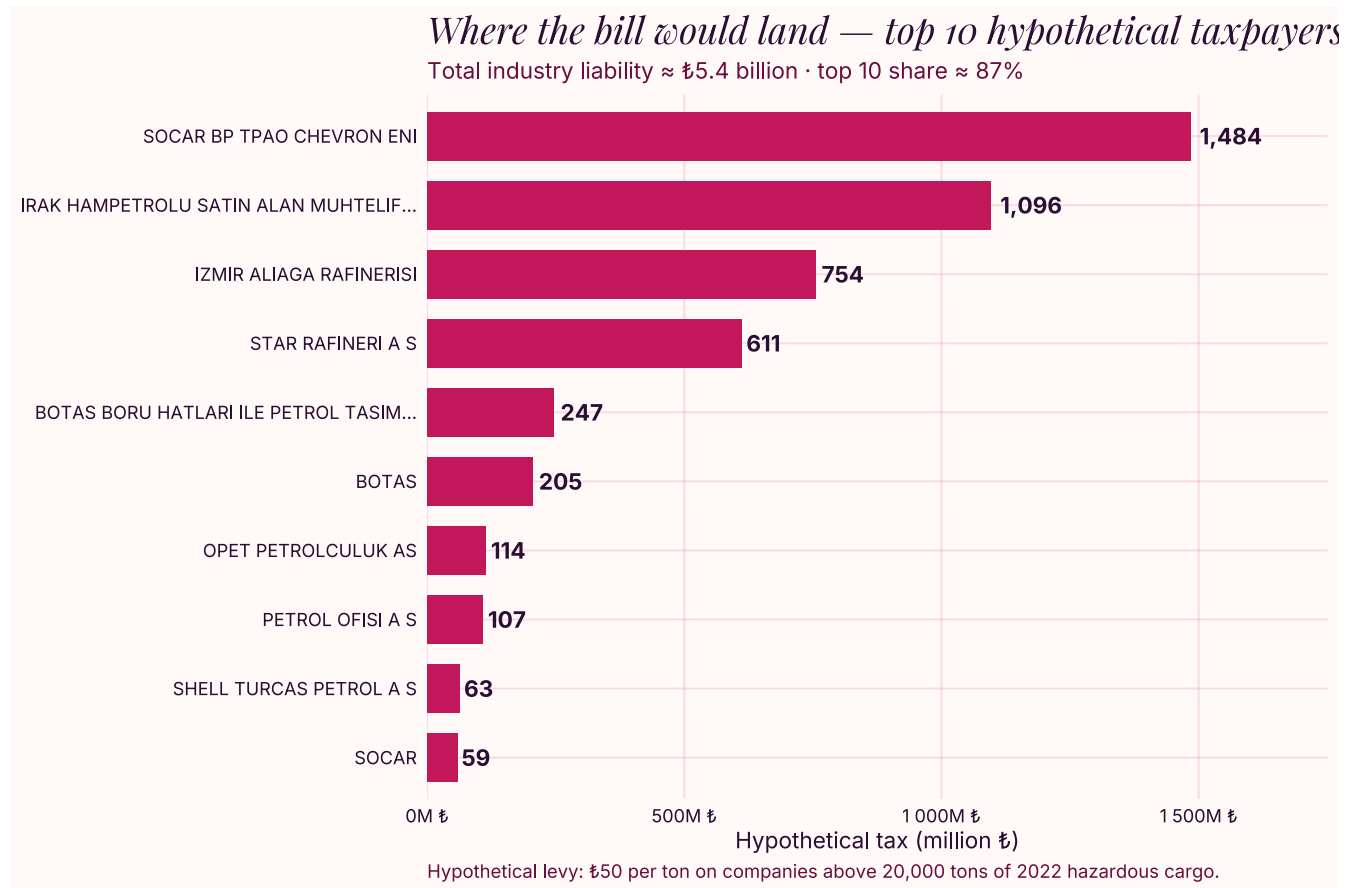


Figure 4: Ten largest hypothetical taxpayers under a 50/ton levy on companies above 20,000 tons.

The top ten alone shoulder the **majority** of the bill. The next several dozen companies above the threshold pay meaningful but much smaller amounts; everyone else pays nothing under this design.

```
top10_tax |>
  transmute(
    Company = label_short,
    `Total tonnage` = scales::label_number(big.mark = ",", accuracy = 1)(total_mt),
    `Hypothetical tax ( million)` = scales::label_number(big.mark = ",", accuracy = 1)(tax_TL / 1e6)
  ) |>
  kable() |>
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 1: Top 10 hypothetical taxpayers (50/ton on companies above 20,000 t).

Company	Total tonnage	Hypothetical tax (million)
SOCAR BP TPAO CHEVRON ENI	29,672,726	1,484
IRAK HAMPETROLU SATIN ALAN MUHTELIF...	21,922,189	1,096
IZMIR ALIAGA RAFINERISI	15,088,213	754
STAR RAFINERI A S	12,212,641	611
BOTAS BORU HATLARI ILE PETROL TASIM...	4,937,501	247
BOTAS	4,090,407	205
OPET PETROLÇULUK AS	2,276,341	114
PETROL OFISI A S	2,141,034	107
SHELL TURCAS PETROL A S	1,265,607	63
SOCAR	1,173,424	59

Sensitivity to the threshold

What if the threshold weren't 20,000 tons? The plot below traces how the **number of taxpayers** and the **share of tonnage taxed** move as we slide it.

```
thresholds <- c(5e3, 1e4, 2e4, 5e4, 1e5, 2.5e5, 5e5, 1e6)

sens <- tibble(threshold = thresholds) |>
  rowwise() |>
  mutate(
    n_taxed      = sum(company_totals$total_mt > threshold),
    share_mt     = sum(company_totals$total_mt[company_totals$total_mt > threshold]) /
                  sum(company_totals$total_mt) * 100
  ) |>
  ungroup()

ggplot(sens, aes(x = threshold)) +
  geom_line(aes(y = share_mt), color = "#C2185B", linewidth = 1.2) +
  geom_point(aes(y = share_mt), color = "#C2185B", size = 3) +
  geom_text(
    aes(y = share_mt, label = sprintf("n=%d", n_taxed)),
    vjust = -1, color = "#00838F", fontface = "bold", size = 3.4, family = "Inter"
  ) +
  scale_x_log10(labels = scales::label_number(big.mark = ",", scale_cut = scales::cut_short_scale())) +
  scale_y_continuous(limits = c(0, 105), labels = function(x) paste0(x, "%")) +
  labs(
    title      = "How sharp is the cliff at 20,000 tons?",
    subtitle   = "Even a 5x or 10x stricter threshold barely loses tonnage coverage - the giants dominate.",
    x          = "Threshold (metric tons, log scale)",
    y          = "Share of total tonnage above threshold",
    caption    = "Annotations show the number of companies caught at each threshold."
  ) +
  theme_riyo()
```

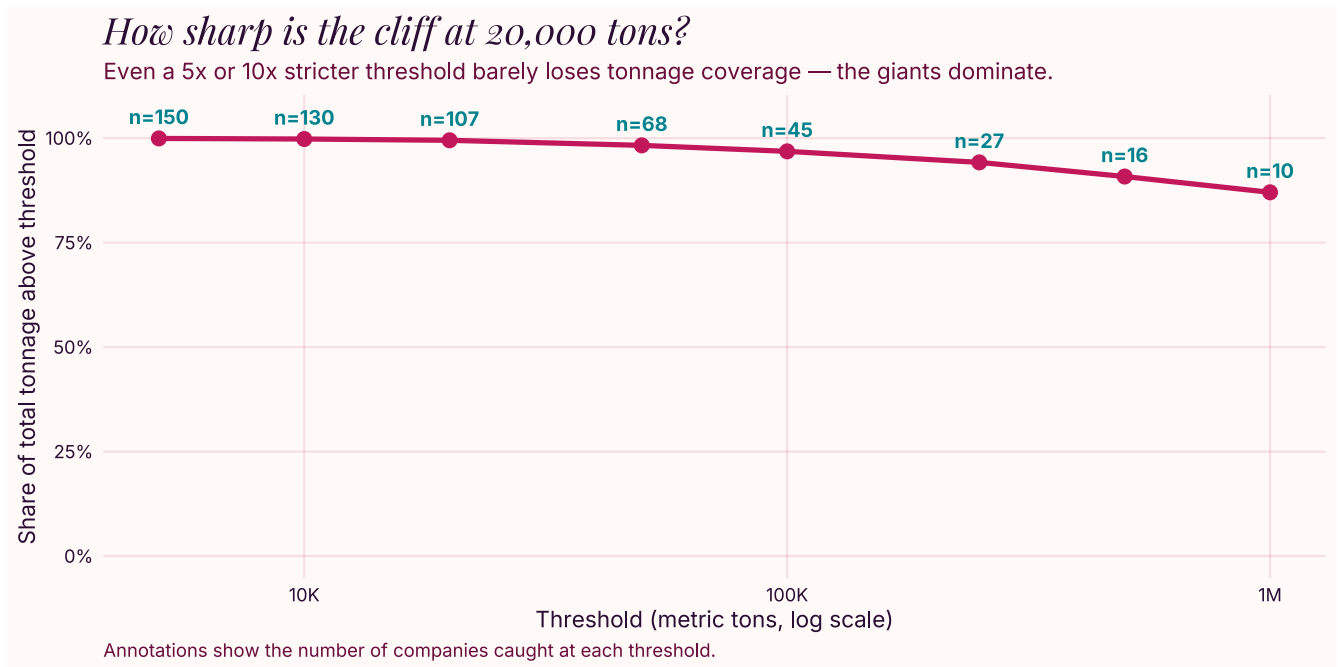


Figure 5: Sensitivity of the policy to the threshold value.

The take-away: because the upper tail is so heavy, **moving the threshold up by an order of magnitude barely reduces tonnage coverage** — but it sharply reduces the *number* of companies subject to the policy. That's a useful lever for designers who want a high-impact, low-friction policy.

5. Where do these companies handle their cargo? — port-authority breakdown

```
port_totals <- hns2022 |>
  group_by(port_authority_short) |>
  summarise(total_mt = sum(cargo_mt), .groups = "drop") |>
  arrange(desc(total_mt))

ggplot(port_totals, aes(x = fct_reorder(port_authority_short, total_mt), y = total_mt / 1e6)) +
  geom_col(fill = "#FF2D87", width = 0.65) +
  coord_flip() +
  scale_y_continuous(labels = scales::label_number(suffix = " M ton")) +
  labs(
    title = "Aliağa and Ceyhan run away with it.",
    subtitle = "Total hazardous cargo handled in 2022, by regional port authority.",
    x = NULL, y = "Total cargo (million metric tons)"
  ) +
  theme_riyo()
```

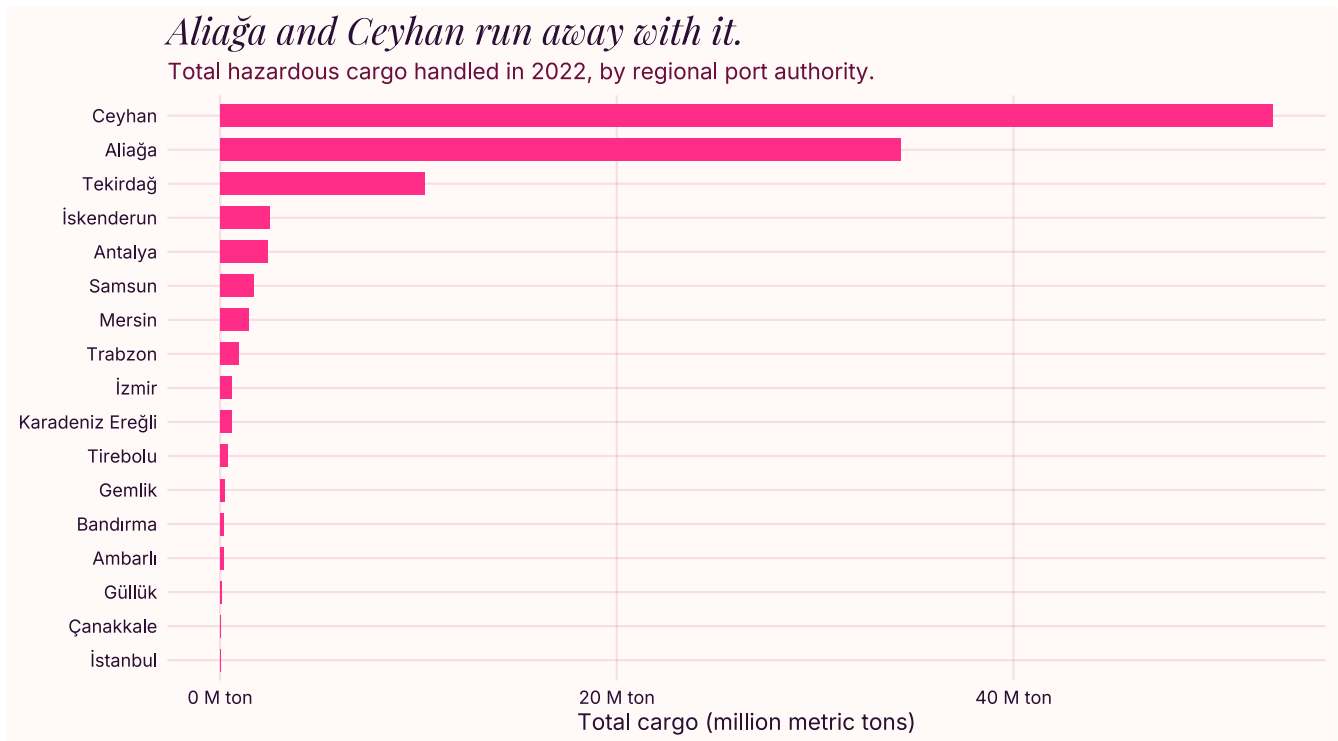


Figure 6: Total hazardous cargo by port authority, 2022.

6. By coastal region: where does the spread sit?

Aggregating to province level can let outliers (a single mega-refinery, a single LPG terminal) dominate the picture. Grouping ports by Türkiye's four coastal regions gives a more balanced view of how *typical* hazardous cargo movements compare across the country, alongside the giants. The box plot below shows every leaf record in the dataset, plotted on a logarithmic axis to keep both small and very large shipments visible. The box itself shows the middle 50% of cargo movements, the line inside the box is the median, and the dots above the whiskers are outliers, often individual mega-shipments.

```
region_map <- tibble::tribble(
  ~province,    ~region,
  "Balıkesir",  "Marmara",
  "Tekirdağ",   "Marmara",
  "İstanbul",   "Marmara",
  "Çanakkale",  "Marmara",
  "Bursa",      "Marmara",
  "İzmir",      "Aegean",
  "Muğla",      "Aegean",
  "Antalya",    "Mediterranean",
  "Mersin",     "Mediterranean",
  "Adana",      "Mediterranean",
  "Hatay",      "Mediterranean",
  "Samsun",     "Black Sea",
  "Trabzon",    "Black Sea",
  "Giresun",    "Black Sea",
  "Zonguldak",  "Black Sea"
)

hns_region <- hns2022 |>
  left_join(region_map, by = "province") |>
  mutate(region = factor(region,
```

```

      levels = c("Marmara", "Aegean", "Mediterranean", "Black Sea"))

region_palette <- c(
  "Marmara"      = "#C2185B",
  "Aegean"       = "#FF2D87",
  "Mediterranean" = "#E91E63",
  "Black Sea"    = "#880E4F"
)

ggplot(hns_region, aes(x = region, y = cargo_mt, fill = region)) +
  geom_boxplot(alpha = 0.85, color = "#3D0823", outlier.color = "#3D0823",
    outlier.size = 1.8, width = 0.55) +
  scale_y_log10(labels = scales::label_number(scale_cut = scales::cut_short_scale())) +
  scale_fill_manual(values = region_palette, guide = "none") +
  labs(
    title = "Cargo tonnage per record, by coastal region",
    subtitle = "Each point is one company at one facility in 2022. Log-10 axis.",
    x = NULL,
    y = "Cargo per record (metric tons, log scale)",
    caption = "Box: middle 50%. Line: median. Whiskers: 1.5x IQR. Dots: outliers."
  ) +
  theme_riyo()

```

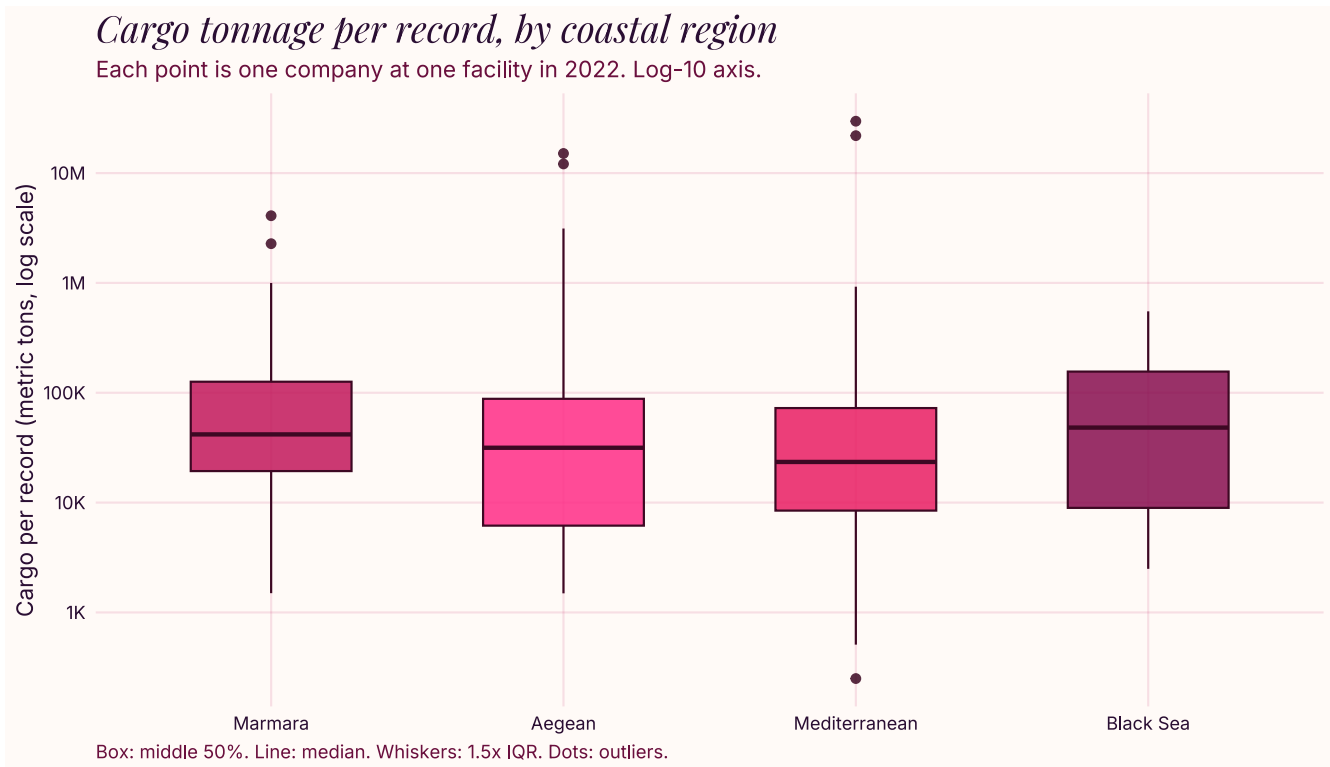


Figure 7: Distribution of leaf-record cargo tonnage across Türkiye's four coastal regions, 2022 (log-10 axis).

The four regions tell different stories. Marmara and the Black Sea share the highest *typical* (median) cargo size per record, both sitting around the 50,000-ton mark, reflecting the steady volume of LPG and petroleum terminals around İstanbul, Tekirdağ, Samsun and Karadeniz Ereğli. The Mediterranean has the lowest median yet hosts the most extreme outliers, driven by a handful of mega-shipments at Ceyhan and İskenderun that sit far above the rest of the region. The Aegean's median falls in the middle, but its outliers (the Aliğa refinery and Star Aliğa terminal) are among the largest single records in the entire dataset. Across all four regions, the gap between an

average shipment and the regional giant spans three to four orders of magnitude, which is the recurring shape of this dataset: a steady base of routine handling on top of a thin layer of exceptional volume.

7. Conclusions

- **Geography:** Türkiye’s hazardous cargo handling is **not evenly distributed**. A small set of provinces, driven by refining and large export terminals, dominates national volume.
- **Industry structure:** The market is a **classic heavy-tailed distribution**. Many small handlers, a few giants moving most of the tonnage.
- **Policy implication:** Because volume is so concentrated, a tonnage-based levy is **administratively narrow** — it would touch a minority of companies but capture most of the activity. That makes it both **easy to enforce** and **politically charged**.
- **Sensitivity:** Moving the threshold from 20,000 to 100,000 or 200,000 tons would *not* meaningfully reduce coverage of total tonnage but would *significantly* reduce the number of regulated firms — a real lever for policy designers.

Visit our [Data page](#) for documentation and the cleaned dataset, and the [Team page](#) to meet the people behind it.